# $C^3$: Committed Confidence Calibration Selection Bias in Confidence-Based Token Commitment in Diffusion Language Models

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## Abstract

Masked diffusion language models commit, at each step, the positions whose top token is most probable, so confidence picks the tokens it then describes; yet calibration is audited over random masks. On teacher-forced traces from LLaDA-8B, Dream-7B and a 110M model of our own, we audit calibration conditional on this selection (C1). In the states a confidence decoder visits, the gap between a committed token's confidence and its accuracy grows with the number of candidates it beat in all seven model and task arms (Holm-corrected, held-out documents), and on the tokens it commits while 32–63 positions are still masked, confidence exceeds accuracy by 1.8 to 4.4 times the random-mask calibration error (descriptive). An exact split of the gap isolates a field term, the part that is a winner's curse: in those states it grows for both instruct models on both tasks, and for the base models it is near the estimator's bias. C2 computes the dependency and estimation errors of committing several tokens at once exactly on a JSON probe with known joint distribution: dependency dominates only for strongly coupled tokens committed several at a time, and on real text a proxy for it is near zero. C3 commits every position whose recalibrated confidence is at least $1 - \alpha$, with no extra forward pass. In an exploratory matched-speed comparison on GSM8K it beats decoders that commit a fixed number of tokens per step by 5 to 8 percentage points, but not a raw confidence threshold: recalibration gives the cutoff a meaning, not extra accuracy.

## 1 Introduction

A masked diffusion language model writes by unmasking. Each step is one forward pass; the decoder then commits some positions for good. LLaDA and Dream commit the $k$ positions they are most confident about, and threshold rules such as Fast-dLLM commit every position above a confidence cutoff (Nie et al., 2025; Ye et al., 2025; Chang et al., 2022; Wu et al., 2025). Confidence thus chooses which tokens are committed and is read as the chance they are right, so on committed tokens it is a selected quantity. The largest of many noisy estimates tends to be one whose error pushed it up, the winner's curse (Andrews et al., 2024), and a commit out of many masked positions is such a maximum.

Calibration of diffusion LMs, where measured, is over whole answers (Yadav et al., 2026; Jiang and Park, 2026), not per committed token. Nor is the bias automatic: selection does not bias a correct posterior (Dawid, 1994; Efron, 2011). We show that the gap between confidence and accuracy on committed tokens splits exactly, when positions' errors are independent given the state, into ordinary miscalibration and a *field term*, nonzero only if accuracy at fixed confidence depends on context (Proposition 2, §2). Only the field term is a winner's curse, and only an audit at the decoder's selection can see it.

Three contributions follow.

- **C1, a selection-conditional calibration audit** (§4). On teacher-forced traces from LLaDA-8B, Dream-7B (Base and Instruct) and a 110M model we trained, we measure the committed-token gap and its field term against how many candidates a token beat, with Holm-corrected tests on held-out documents. In the states a confidence decoder visits, the gap grows with field size for every model (in random masked states its sign varies). In those states its field term grows too for both instruct models on both tasks; for the base and 110M models it is of the order of the estimator's bias. At equal confidence, tokens are right less often in those states than in random masks of the same size, in every model (exploratory).

- **C2, an instrument for the two errors of parallel commitment** (§5). Committing several tokens in one step incurs a dependency error (A) and an estimation error (B) (Eq. 1), computed exactly on a JSON probe with known joint distribution. (A) overtakes (B) only for strongly coupled slots ($\rho \geq 0.5$) and several tokens per step; confidence ranking raises (A), not (B); and at $\rho = 1$ (B) falls as (A) rises along $k$. On real text a proxy for (A) is near zero at 2–8 tokens per step.

- **C3, a calibrated commitment policy** (§6). C3a commits as many positions as have recalibrated confidence at least $1 - \alpha$, in the model's own order, with no extra forward pass. At matched speed (LLaDA-8B-Instruct, GSM8K) it beats top-$k$, margin and PC-Sampler but not the Fast-dLLM threshold it recalibrates (exploratory).

## 2 Setup and theory

**Masked diffusion decoding.** A masked diffusion LM (Austin et al., 2021a; Sahoo et al., 2024; Nie et al., 2025; Ye et al., 2025) writes by unmasking: in a *state* $s$ with masked set $M$, one forward pass gives at each $i \in M$ a distribution $p_{\theta,i}$ given the observed tokens $x_{\text{obs}}$, with top-1 token $v_i$, *confidence* $c_i = p_{\theta,i}(v_i)$ and log-odds $\ell_i = \log \frac{c_i}{1-c_i}$. The decoder *commits* positions for good: the $k$ most confident per step (Chang et al., 2022) or all above a cutoff (Wu et al., 2025). Committing $S \subseteq M$ in one step draws its tokens from the product of their marginals; with $p^*$ the true joint of $x_S$ given $x_{\text{obs}}$ and $p_i^*$ its marginals,

$$\mathrm{KL}\big(p^* \,\|\, \textstyle\prod_{i\in S} p_{\theta,i}\big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})}_{\text{(B) estimation}}, \quad (1)$$

where $\mathrm{TC}(x_S \mid x_{\text{obs}}) = \sum_{i\in S} H(p_i^*) - H(p_S^*)$ is the total correlation, zero iff they are conditionally independent. (A) is what no product of marginals can represent, (B) the marginals' error (cf. Ben-Hamu et al., 2025).

**The committed-token gap.** Under teacher forcing (§3) correctness $a_i = \mathbf{1}\{v_i = x_{0,i}\}$ against the true document $x_0$ is observed. A commit is a selection, so we model a field of $m$ candidates as a random pool $P \subseteq M$, won by $i = \arg\max_{j\in P} \ell_j$ (ties broken at random). With $\mathbb{E}_m$ the average over states, pools of size $m$ and winners, the *committed-token gap* is

$$\Delta(m) = \mathbb{E}_m\,[\,c - a\,], \quad (2)$$

confidence minus accuracy, so positive means overconfident. $\Delta(1)$ is the ordinary signed calibration error of a masked position; larger $m$ stands for a larger field, as at early steps and in long outputs, and random pools vary $m$ with the state held fixed.

**The winner's curse.** If the winner's curse (Efron, 2011; Andrews et al., 2024) acts on decoding, $\Delta(m)$ grows with $m$. Let $\pi(\ell) = \Pr(a = 1 \mid \ell)$ be the calibration curve over all masked positions (the $m = 1$ population) and $\pi(\ell, s) = \Pr(a = 1 \mid \ell, s)$ its context-aware version.

**Proposition 1 (Winning adds nothing)** *Fix $s$. Suppose that, given $s$, each pair $(\ell_i, a_i)$ is independent of the other positions' scores $(\ell_j)_{j\neq i}$, and that the pool and the tie-breaks are drawn independently of all $(\ell_j, a_j)_{j\in M}$. Then the winner $i$ of $P$ satisfies*

$$\Pr\big(a_i{=}1 \;\big|\; \ell_i,\, s,\, i = \arg\max_{j\in P} \ell_j\big) = \pi(\ell_i, s).$$

Winning depends only on the other scores, which say nothing about $a_i$, so selection does not bias a correct posterior (Dawid, 1994; Efron, 2011); it acts only through context:

**Proposition 2 (Exact split of the gap)** *Under the conditions of Proposition 1,*

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\textit{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\textit{field term}}, \quad (3)$$

*and the field term is exactly zero at $m = 1$.*

Proofs are in Appendix A.3. Own miscalibration, reweighted towards where winners sit, can make $\Delta(m)$ grow with $m$ with no winner's curse; the *field term*, nonzero only if accuracy at fixed confidence depends on context (it requires a nonzero grouping loss; Kull and Flach, 2015; Perez-Lebel et al., 2023), is the winner's curse in a testable form. If positions of a state share errors, Proposition 1 fails and (3) gains a third term, $\mathbb{E}_m[\pi(\ell, s) - a]$ (§3).

| Model | Text | Prec. | Docs |
|---|---|---|---|
| LLaDA-8B-Base | FineWeb-Edu | fp16 | 1,979 |
| Dream-7B-Base | FineWeb-Edu | fp16 | 1,976 |
| Our 110M MDLM | FineWeb-Edu | fp32 | 1,979 |
| LLaDA-8B-Instruct | GSM8K | fp16 | 1,319 |
| LLaDA-8B-Instruct | MBPP | fp16 | 499 |
| Dream-7B-Instruct | GSM8K | fp16 | 1,319 |
| Dream-7B-Instruct | MBPP | fp16 | 499 |

Table 1: C1 trace runs: the seven primary arms. Also run: the time-split set (500 documents for each base model and the 110M model, same precision) and LLaDA-8B-Base in 4-bit on 198 FineWeb-Edu documents. In all, 0.49B position rows, 208 GPU-hours.

# 3 Measurement instrument

## 3.1 Models and data

**Models.** We audit LLaDA-8B (Nie et al., 2025) and Dream-7B (Ye et al., 2025), Base and Instruct, in fp16. As neither discloses its pretraining data, we also trained a 110M MDLM (Sahoo et al., 2024) on data we control: a bidirectional transformer (12 layers, width 768, 12 heads, GPT-2 BPE plus a mask token, length 256) trained on 10.0B tokens (169,055 steps) of FineWeb-Edu sample-10BT (Penedo et al., 2024). Its training stream excludes the calibration and reserve documents and matches no time-split document.

**Data.** Base and 110M models read 2,000 FineWeb-Edu documents, reservoir-sampled from the first 200,000 of at least 1,200 characters, cut to 256 tokens per tokeniser; shorter ones are left out. Instruct models read the GSM8K (Cobbe et al., 2021) and MBPP (Austin et al., 2021b) test sets with the question visible and only the reference answer maskable; one MBPP document over 1,024 tokens is excluded. A time-split set of 500 English Wikipedia articles written after LLaDA and Dream were released, unseen by every model, is a robustness check. A document is contaminated if a 7–8B Base model, given its first 50 tokens, greedily reproduces a run of $\geq$ 32 of the next 50 (2/1,979 LLaDA, 1/1,976 Dream); LLaDA and Dream FineWeb endpoints are also reported without them. Every set is split 50/50 by document with one fixed seed (FineWeb: 1,000/1,000 before the length filter). Calibrators, the spline $f$ of Eqs. 4–5, knots and temperatures are fitted on the fit half; every C1 number comes from the report half. 500 separate reserve documents served the pilot and the non-determinism floor.

## 3.2 Traces

Each measurement is one forward pass on a true document with some positions masked, recording at each masked position $v_i$, $c_i$, $\ell_i$, $a_i$ (§2), the true token's probability, entropy, margin, the top-20 and whether it was committed. A document has three kinds of states: *uniform*, each position masked independently at rate $t \in \{0.1, 0.2, \ldots, 0.9\}$, 2 masks per rate; *decoder*, every state of a teacher-forced confidence top-$k$ trajectory, $k \in \{2, 4, 8\}$, which starts fully masked, commits the $k$ most confident positions, fills them with the *true* tokens and repeats (the states a decoder visits if always right); and *matched uniform*, a uniform mask of each decoder state's size, so only the pattern differs: 466 passes per document. In every stored state with at least 32 masked positions we draw 20 random pools of each size $m \in \{1, 2, 4, 8, 16, 32\}$, rank them by $\ell$ with random tie-breaks (fp16 confidences saturate) and keep the top four ranks; $\Delta(1)$ is exact over all masked positions of those states.

## 3.3 Estimands

**Calibrators.** Both are fitted on the fit half. $\hat{\pi}(\ell)$ is isotonic in $\ell$. $\hat{\pi}(\ell, s)$ is logistic in a restricted cubic spline of $\ell$ plus the state's field features (median $\ell$ of its masked positions, mask rate, $\log |M|$, decoding step) and their products with $\ell$, so the slope varies too; an additive form, without these products, is a sensitivity check. Documents weigh equally (Appendix A.2). The calibrators are fitted on every state but the averages use only states with at least 32 masked positions, so the estimated field term need not be zero at $m = 1$: E2 is a contrast.

**E1, E2 and the native gap.** E1 is $\Delta(m)$ (2), with primary contrast $\Delta(32) - \Delta(1)$, beside 15-bin equal-mass ECE over all masked positions. E2 is the field term $\mathbb{E}_m[\hat{\pi}(\ell) - \hat{\pi}(\ell, s)]$ of (3), with primary contrast field$(32)-$field$(1)$. On the same winners, own miscalibration $\mathbb{E}_m[c-\hat{\pi}(\ell)]$, the field term and $\mathbb{E}_m[\hat{\pi}(\ell, s) - a]$ sum to $\Delta$ exactly; the last, zero under Proposition 1, is a specification check, as is the intra-class correlation of residuals $a - \hat{\pi}(\ell, s)$ within states. The *native* gap is $c - a$ over the tokens the decoder commits, grouped by $|M_t|$ at commit time.

**Rank diagnostics.** Eq. 4 is on pool winners, Eq. 5 on the top four ranks; $\eta_i = \alpha\,[+\alpha_s] + f(\ell_i)$:

$$\Pr(a_i{=}1) = \sigma\big(\eta_i + \beta_z z_{r,m}\big), \tag{4}$$

$$\Pr(a_i{=}1) = \sigma\big(\eta_i + \textstyle\sum_{k=2}^{4} \gamma_k \mathbf{1}\{r{=}k\}\big), \tag{5}$$

where $z_{r,m}$ is the expected $r$-th largest of $m$ standard normals and $f$ a restricted cubic spline with 9 knots, fitted on the fit half and carried over as a fixed offset. Each is fitted without and with state fixed effects $\alpha_s$, which absorb the field term because it lives between states: $\beta_z < 0$ without them and $\approx 0$ with them points to a field effect, $\beta_z \neq 0$ with them to shared errors; the reading is approximate (§3.5), so $\beta_z$ is a diagnostic.

**E3 and controls.** E3, a shift in accuracy between decoder and uniform states at equal confidence and mask rate, and controls for position, token type, temperature and precision are defined in Appendix A.2.

### 3.4 Inference

**Bootstrap.** The document is the unit. Intervals are 95% percentile intervals over 10,000 document resamples, one matrix per run shared across pool sizes, estimands and arms, so contrasts come from differenced draws. Eqs. 4–5 are refit per resample and use 1,000 (knot runs: 200). Calibrators are not refitted, so E2 intervals condition on the fit half.

**Declaring effects, multiple testing.** An effect is declared only if its interval excludes 0 *and* its magnitude exceeds twice the non-determinism floor, the largest change in $\Delta(m)$ over five repeats of 20 reserve documents per model family. Repeats were bitwise identical, so every floor is 0 and the interval decides. Holm's correction (two-sided bootstrap $p$-values) runs within families: the E1 contrasts and the E2 contrasts, each over the seven arms of Table 1 in uniform and $k{=}4$ decoder states (14 tests each), and $\gamma_2, \gamma_3, \gamma_4$ within each arm and kind of state.

### 3.5 Validation

**Known truth.** We first ran the pipeline on simulated traces in four worlds: calibrated everywhere; miscalibrated alike in every context; miscalibrated by context (the winner's curse of Proposition 2); and errors shared within a state (violating Proposition 1). The E1 contrast was zero only in the first, and $\beta_z$ without fixed effects strongly negative only in the third. Three findings shape §4: the field-term estimator reads high (at $m = 32$, 0.001 and 0.002 where the truth is zero; above the truth otherwise); fixed effects do not fully clean $\beta_z$ in the third; and shared errors give a *negative* field term with a *positive* $\beta_z$ without fixed effects. Too few knots let curvature load onto $z_{r,m}$; we use 9.

## 4 Selection and confidence (C1)

C1 asks whether a committed token's confidence overstates its accuracy more when it won a larger field, and how much of that is the field term of Proposition 2. Every number is from the report half (§3).

**Arms and claims.** We claim an E1 or E2 effect only if it is declared *and* Holm rejects it at $\alpha = 0.05$ in its family: the seven arms in uniform and $k{=}4$ decoder states (§3.4; a missing cell counts as $p = 1$). Everything else is exploratory, reported with its interval and no claim: decoder states at $k \in \{2, 8\}$, matched uniform states, $\beta_z$, $\gamma_k$, E3, the native gap, the controls and the robustness runs (§4.4).

### 4.1 Does the gap grow with the field?

Table 2 gives both contrasts for every arm. Of the 14 E1 tests, Holm rejects 12 and 14 are declared; 12 meet both and are claims. For E2 the counts are 14, 14 and 14. The signs differ, so we do not pool. E1 is positive and claimed in every $k{=}4$ decoder-state cell and in uniform states of Dream-7B-Instruct on both tasks: there a token that wins a field of 32 is more overconfident than a random masked position. It is negative and claimed in uniform states of both base models and of LLaDA-8B-Instruct on GSM8K, where the gap shrinks with the field. The two cells without a claim, uniform states of LLaDA-8B-Instruct on MBPP and of the 110M model, are both declared but not claimed (both $p_{\text{Holm}} = 0.0524$). Sizes differ as much as signs: $|\text{E1}| \leq 0.015$ for the base and 110M models, up to 0.26 for the instruct models on MBPP.

Decoder states change the picture: E1 changes sign for both base models and LLaDA-8B-Instruct on GSM8K, and is claimed only in decoder states for LLaDA-8B-Instruct on MBPP and the 110M model. Uniform states matched to the decoder's mask counts (exploratory) keep the base models' negative sign (LLaDA: $-0.002$ $[-0.003, -0.001]$), so there the change comes from which positions the decoder leaves

| Arm | Masks | E1 | Holm | E2 | Holm | $\beta_z$ | $\beta_z^{\mathrm{FE}}$ |
|---|---|---|---|---|---|---|---|
| LLaDA Base | uniform | $-0.003^{*,\dagger}$ [$-0.004$, $-0.002$] | yes | $0.0010^{*,\dagger}$ [0.0009, 0.0010] | yes | $-0.03^{\ddagger}$ | $-0.01^{\ddagger}$ |
| | dec. $k$=4 | $0.006^{*,\dagger}$ [0.005, 0.008] | yes | $0.0036^{*,\dagger}$ [0.0035, 0.0037] | yes | $0.01^{\ddagger}$ | 0.01 |
| Dream Base | uniform | $-0.002^{*,\dagger}$ [$-0.004$, $-0.001$] | yes | $0.0011^{*,\dagger}$ [0.0011, 0.0012] | yes | $-0.03^{\ddagger}$ | 0.00 |
| | dec. $k$=4 | $0.015^{*,\dagger}$ [0.013, 0.017] | yes | $0.0029^{*,\dagger}$ [0.0028, 0.0029] | yes | $-0.02^{\ddagger}$ | $-0.02^{\ddagger}$ |
| LLaDA-I GSM8K | uniform | $-0.022^{*,\dagger}$ [$-0.025$, $-0.018$] | yes | $-0.003^{*,\dagger}$ [$-0.004$, $-0.003$] | yes | $-0.16^{\ddagger}$ | $-0.02$ |
| | dec. $k$=4 | $0.024^{*,\dagger}$ [0.016, 0.032] | yes | $0.021^{*,\dagger}$ [0.020, 0.023] | yes | $-0.10^{\ddagger}$ | $-0.02$ |
| LLaDA-I MBPP | uniform | $0.025^{\dagger}$ [0.003, 0.051] | no | $0.013^{*,\dagger}$ [0.010, 0.016] | yes | $-0.56^{\ddagger}$ | $-0.23^{\ddagger}$ |
| | dec. $k$=4 | $0.156^{*,\dagger}$ [0.126, 0.189] | yes | $0.036^{*,\dagger}$ [0.031, 0.041] | yes | $-0.04^{\ddagger}$ | 0.01 |
| Dream-I GSM8K | uniform | $0.061^{*,\dagger}$ [0.052, 0.070] | yes | $0.021^{*,\dagger}$ [0.020, 0.023] | yes | $-0.42^{\ddagger}$ | $-0.03$ |
| | dec. $k$=4 | $0.072^{*,\dagger}$ [0.064, 0.081] | yes | $0.038^{*,\dagger}$ [0.036, 0.039] | yes | $-0.05^{\ddagger}$ | $0.02^{\ddagger}$ |
| Dream-I MBPP | uniform | $0.257^{*,\dagger}$ [0.227, 0.290] | yes | $0.176^{*,\dagger}$ [0.159, 0.194] | yes | $-0.76^{\ddagger}$ | $-0.07$ |
| | dec. $k$=4 | $0.186^{*,\dagger}$ [0.151, 0.224] | yes | $0.076^{*,\dagger}$ [0.066, 0.086] | yes | $-0.19^{\ddagger}$ | 0.01 |
| MDLM-110M | uniform | $0.00152^{\dagger}$ [0.00003, 0.00301] | no | $0.0019^{*,\dagger}$ [0.0018, 0.0019] | yes | $-0.01^{\ddagger}$ | 0.00 |
| | dec. $k$=4 | $0.009^{*,\dagger}$ [0.007, 0.011] | yes | $-0.0023^{*,\dagger}$ [$-0.0023$, $-0.0022$] | yes | $0.02^{\ddagger}$ | 0.01 |

Table 2: C1 primary endpoints. E1 = $\Delta(32) - \Delta(1)$ and E2 = field(32)$-$field(1), with 95% intervals (10,000 document resamples; $\beta_z$: 1,000). Holm: decision of the Holm step. $^{\dagger}$declared; $^{*}$claimed. $\beta_z$, $\beta_z^{\mathrm{FE}}$: Eq. 4 without and with state fixed effects ($^{\ddagger}$interval excludes 0; $^{\S}$no convergence). —: absent. Time-split rows: code archive.

$^{*}$ Claimed: declared (interval excludes 0 and effect above twice the floor) and rejected by Holm at $\alpha = 0.05$ within its family of 14 cells per contrast; a cell without results counts as $p = 1$. Missing: none.

masked; for LLaDA-8B-Instruct on GSM8K they are already positive (0.016 [0.010, 0.022]), so there it comes partly from how many.

**How much is the field term?** At $m = 32$ LLaDA-8B-Base's uniform-state gap splits into own miscalibration 0.002, field term 0.001 and a residual; E2 is 0.0010 [0.0009, 0.0010] (decoder states: 0.0036 [0.0035, 0.0037]). E2 is claimed in all 14 cells and positive in 12. For the base and 110M models every |E2| is below 0.005, of the order of what the estimator read on simulated traces with no field term (§3.5), so we do not read these as a winner's curse. In seven of the eight instruct cells E2 is 0.013 to 0.176 (Dream-7B-Instruct, MBPP, uniform states, where E1 is 0.257): there the field term grows with $m$, as a winner's curse predicts: at equal confidence, winners of large fields come from contexts where tokens are right less often. The eighth, LLaDA-8B-Instruct on GSM8K in uniform states, is negative. E2 intervals condition on the fitted calibrators (§3.4). Positions of a state share errors in every arm (within-state ICC of residuals above 0 throughout: at most 0.006 in the base and 110M models, 0.036 to 0.056 in the instruct models), so the split has its third term, $\mathbb{E}_m[\hat{\pi}(\ell, s) - a]$. At $m = 32$ it is near 0 in the base and 110M models except LLaDA-8B-Base decoder states ($-0.008$ [$-0.009$, $-0.006$]), and 0.055 to 0.103 on MBPP and in uniform states of Dream-7B-Instruct on GSM8K, larger than E2 in four of these five cells; there E2 and $\beta_z$ are read with that in mind. The simulated shared-error world gave a field term of $-0.007$ [$-0.009$, $-0.005$] with a negative third term, unlike these cells; this does not rule shared errors out (full table in the code archive).

### 4.2 The selection diagnostic $\beta_z$

$\beta_z$ (Eq. 4) asks whether winning a larger field predicts correctness beyond the winner's own log-odds; shared errors gave a *positive* $\beta_z$ without fixed effects in simulation (§3.5), so we read it beside the sign of E2. Without fixed effects it excludes 0 in all 14 cells (Table 2) and is negative in 12, and with them its interval holds 0 in 8 of those 12: the pattern of a field effect (small in the base and 110M models, as E2 is); E2 is positive in all of them but uniform states of LLaDA-8B-Instruct on GSM8K. The two positive cells, decoder states of LLaDA-8B-Base and of the 110M model, have E2 near the estimator's bias (the 110M model's negative, the sign simulated shared errors gave). Holm, run within each arm and kind of state, rejects 24 of the 42 tests of $\gamma_2, \gamma_3, \gamma_4$ (Eq. 5), in 8 of the 14 rows (code archive), all positive and all in the instruct models and Dream-7B-Base decoder states: within a state, at equal confidence, lower-ranked pool members are right more often than the win-

ner, which Proposition 1 excludes when its conditions hold. Like the ICC, we read this as a sign of shared errors (or of a calibration slope that varies by state); none is a claim.

### 4.3 Decoder states

E3 compares accuracy at equal confidence and mask rate in decoder and matched uniform states (Appendix A.2); the decoder is teacher-forced, so a difference comes from which positions remain masked, not from its own mistakes. It is positive, its interval excluding 0, in every arm and at every $k \in \{2, 4, 8\}$ (sup-$t$ $p \leq 0.002$; exploratory, full table in the code archive): tokens are right less often in the states a decoder visits, by 0.001 to 0.006 in the base and 110M models and 0.03 to 0.06 in the instruct models. A calibration curve fitted on random masks of the same mask rate therefore overstates accuracy in decoder states by about this much; C3a's is fitted on decoder states (§6). The tokens a decoder commits are themselves more overconfident than random masks suggest: with 32–63 positions masked, $c-a$ on the $k$=4 decoder's commits is 1.8 to 4.4 times its arm's random-mask ECE in every arm, its interval above it (descriptive; Appendix A.2).

### 4.4 Controls and robustness

Position, token type and a temperature per mask-rate bin leave decoder-state E1 with its sign in every arm, but not LLaDA-8B-Base's small uniform-state E1, which token type takes to 0; a 4-bit check does not rule out an effect of precision (Appendix A.2). Five checks repeat the contrasts outside the Holm families (full tables in the code archive). *Contamination:* without the 2 LLaDA and 1 Dream documents the screen flagged (§3.1), both base models' E1 equal their main values. *Time split:* on unseen Wikipedia articles (a quarter of the documents) the small negative uniform-state E1 of the base models is not reproduced (LLaDA-8B-Base: $-0.001$ $[-0.003, 0.002]$): seen text, the change of domain, or too small to detect here; the 110M model's grows (0.012); it saw neither set, so domain, not contamination, moves it. Decoder-state E1 stays positive, its interval excluding 0, for all three. *Knots:* E2 keeps its sign and declared flag with 5 and 13 knots in 14 of 14 cells. *Calibrator form:* with the additive calibrator (§3.3) it does in 12 of 14, though the seven positive instruct-model E2s shrink under it, to 0.000–0.098; the two others, uniform states of LLaDA-8B-Instruct on MBPP (interval now holding 0) and 110M decoder states, are claimed cells whose E2 we therefore do not rely on. *Independent estimator:* an exact all-subsets estimator of $\Delta(m)$ with its own spline-logistic calibrator and own/field split (after Dorigo 2026) reproduces E1 in all four base cells (LLaDA, Dream; decoder $k$=4: 0.007, 0.015; uniform: $-0.003$, $-0.003$), the decoder growth mostly in the own term. Its field terms match E2 in sign and size (no larger than the 0.003 it reads with no field term), except LLaDA's in decoder states ($-0.0004$ vs 0.004), which is not identified (residual at $m = 32$ $-0.005$ $[-0.007, -0.004]$, refits of opposite sign).

## 5 Two sources of error (C2)

Work on parallel decoding targets the dependency term (A) of Eq. 1 (§2) and takes the marginals as accurate (Kim et al., 2026; Ringel et al., 2026; Azangulov et al., 2026; Kang et al., 2026).

### 5.1 A JSON probe with known $p^*$

The probe fills the simple fields of 200 SchemaStore schemas (SchemaStore contributors, 2026), 6–16 per schema, with single-token values; we call them slots. Each slot has a Dirichlet(1) marginal; slots are paired: with probability $\rho$, stated in a marker at the start of the instance, the follower is a fixed function of its leader's value, otherwise an independent draw, so $p^*$ is known exactly. We fine-tune our 110M MDLM (§3) on 200k instances and evaluate 1,000 (200 per $\rho \in \{0, .25, .5, .75, 1\}$); the main model is the fine-tune that never saw the marker masked (Appendix A.1). Decoding starts with every value masked; each step picks $k \in \{1, 2, 4, 8\}$ slots, the most confident or at random, scores (A) and (B) ($p_{\theta,i}$ is the model's probability of each allowed value; mass off the list counts against (B)), and fills them with true values. Terms are summed over steps and divided by the number of slots; bootstrap intervals use 10,000 instance resamples.

### 5.2 (A) overtakes (B) at high $\rho$ and $k$

(A) is zero at $k = 1$ and at $\rho = 0$ and grows with both (Figure 1). For $\rho \geq 0.5$ it overtakes (B) at $k = 8$, or at $\rho = 1$ under confidence top-$k$ already at $k = 4$; each crossover is also where the 95% interval of (A)$-$(B) clears zero. For $\rho \leq 0.25$ (B) dominates up to $k = 8$. At $\rho = 1$, $k = 8$ (confidence), (A) is 0.338 [0.320, 0.355] nats per slot and (B) 0.109 [0.095, 0.123]: 76% of the total.

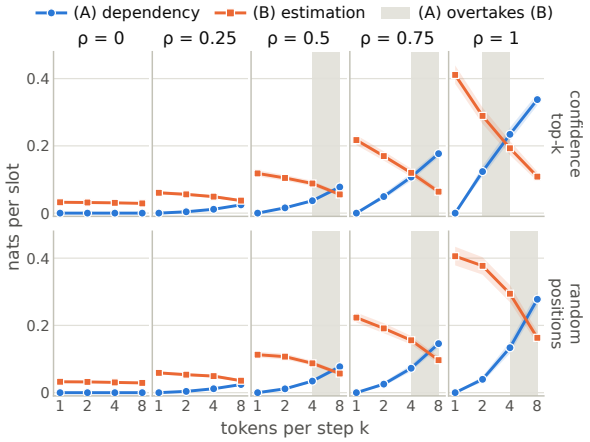



Figure 1: (A) and (B) per slot on the probe (main model), with 95% intervals, against tokens per step $k$. Columns: $\rho$; rows: selection rule. Shaded: the $k$ where (A) exceeds (B) (hatched where the interval of (A)−(B) still holds 0).

**Confidence selection raises (A), not (B).** Where (B) is higher under confidence ranking than at random, the intervals overlap. (A) rises instead ($\rho = 1$, $k = 2$: 0.124 [0.108, 0.140] against 0.040 [0.031, 0.049]): the ranking commits leaders with their followers more often than chance, bringing the crossover forward.

**The terms trade off.** $\rho$ was meant to move only (A), but (B) rises too: at $k = 1$, from 0.032 at $\rho = 0$ to 0.411 at $\rho = 1$. Decoded after its leader, a follower should be a near-copy, but the model reads the marker only in part (Appendix A.1) and stays broad: error in (B). Decoded in the same step, the true marginal is broad too, and the missing coupling counts as (A). At $\rho = 1$, from $k = 1$ to 8, (B) falls to 0.109 while (A) rises from 0 to 0.338. Without fine-tuning, (B) dominates everywhere ((A) is $\leq 4\%$ of the total).

### 5.3 Real text: proxies checked on the probe

The exact values hold for one synthetic distribution and one 110M model; on real text $p^*$ is unknown, so we use proxies. For (A), parallel–sequential disagreement: the fraction of the $k$ most confident masked positions whose token changes if committed one at a time (earlier ranks filled with the model's own tokens) instead of in one pass. For (B), the gap $\Delta = c - a$ and the cross-entropy (CE) of the true token, at the same positions. We run the 110M model and LLaDA-8B-Base on 300 fit-half FineWeb-Edu documents, one state at each mask rate $0.1, \ldots, 1$ (Appendix A.1), and the proxy on the probe.

**The proxy tracks (A) weakly.** Over the 15 confidence-rule cells with $k \geq 2$, its Spearman correlation with the exact (A) is 0.49 [0.28, 0.63] for the main model and 0.28 [0.06, 0.49] for the other. It rises with $k$, barely with $\rho$: at $k = 8$, 0.028 at $\rho = 0$, where (A) is zero, and 0.049 at $\rho = 1$.

**On real text the (A) proxy is near zero.** Over mask rates 0.1–0.9,[1] disagreement at $k = 8$ is 0.0021 for the 110M model and 0.0022 for LLaDA, and at most 0.0003 at $k \leq 4$. CE grows from 0.08 nats at $k = 1$ to 0.40 at $k = 8$ (LLaDA: 0.08 to 0.23), though it includes the text's own entropy and lower ranks are harder positions. $\Delta$ cannot be told from zero at $k = 1$ and is small but positive at $k = 8$: $+0.007$ [0.004, 0.010] and, for LLaDA, $+0.004$ [0.002, 0.007].

So at 2–8 tokens per step the models act on almost no learned dependency between jointly committed positions. The proxy sees only learned dependence, so this does not show that (A) is small; real text gives a direction, not a split in nats.

## 6 A recalibrated commit rule (C3)

**The rule.** A top-$k$ decoder fixes the number of tokens per step in advance; C3a lets calibrated confidence set it. At each step it computes $\hat{c}_i = \hat{\pi}(\ell_i, s)$ with the context-aware calibrator of §3.3 on three field features (median $\ell$, mask rate, $\log|M|$), and commits the $n$ masked positions of the active block with the highest $\ell$, where $n$ is the number with $\hat{c}_i \geq 1 - \alpha$, or one if none qualifies. If $\hat{c}$ is non-decreasing in $\ell$ across a step's positions, these are exactly the positions with $\hat{c}_i \geq 1 - \alpha$, so every committed token except a forced single commit has calibrated error probability $1 - \hat{c}_i \leq \alpha$. The $\ell\times$feature terms do not guarantee this; it holds in a fraction 1.000 of GSM8K and 1.000 of MBPP fit states. $\alpha$ is the one speed knob. C3a changes *how many* tokens are committed, not *which*, and needs no extra forward pass.

**The calibrator.** One calibrator per task is fitted on teacher-forced decoder states ($k \in \{2, 4, 8\}$) of the fit-half LLaDA-8B-Instruct traces (660 GSM8K and 249 MBPP documents) and frozen. A check inside the fit half (not a result) puts its ECE against the raw $\sigma(\ell)$ at 0.006 from 0.061 on GSM8K and 0.007 from 0.039 on MBPP. It is

[1] At rate 1 (all-`[MASK]` input) the 110M model's disagreement over all ten rates is 0.0644 at $k = 8$; LLaDA's is none.

| | GSM8K ($N = 659$) | | | MBPP ($N = 121$) | | |
|---|---|---|---|---|---|---|
| Decoder | tok/step | Acc. [95% CI] | C3a − decoder [95% CI] | tok/step | Acc. [95% CI] | C3a − decoder [95% CI] |
| Top-$k$ | 4.00 | 0.730 [0.695, 0.762] | 0.059 [0.026, 0.093] | — | — | — |
| Threshold | 4.03 | 0.791 [0.758, 0.820] | −0.002 [−0.014, 0.012] | 3.73 | 0.355 [0.276, 0.444] | 0.000 [−0.033, 0.033] |
| Margin | 4.00 | 0.736 [0.701, 0.768] | 0.053 [0.020, 0.088] | — | — | — |
| Entropy | 4.28 | 0.794 [0.761, 0.823] | −0.005 [−0.035, 0.026] | 3.83 | 0.355 [0.276, 0.444] | 0.000 [−0.058, 0.058] |
| KLASS† | 3.97 | 0.505 [0.467, 0.543] | 0.284 [0.243, 0.325] | 3.87 | 0.066 [0.034, 0.125] | 0.289 [0.198, 0.380] |
| PC-Sampler | 4.00 | 0.707 [0.671, 0.741] | 0.082 [0.050, 0.112] | — | — | — |
| C3a | 4.04 | 0.789 [0.756, 0.819] | ref. | 3.61 | 0.355 [0.276, 0.444] | ref. |

Table 3: Decoders at matched speed: speed 2 (target 4 tokens per step), C3a at $\alpha = 0.271$ (GSM8K) and 0.322 (MBPP). Wilson intervals for accuracy, paired bootstrap for C3a − decoder (positive favours C3a). †Off its published operating point (Appendix A.4).

then applied to free generation, which starts from 256 masks; its fit states lie inside a reference answer. Of C3a's steps, 0.017 (GSM8K) and 0.034 (MBPP) have features outside the fitted range.

**Matched speed.** LLaDA-8B-Instruct generates freely on the report halves of GSM8K (659 problems) and MBPP (121): zero-shot chat prompt, 256 tokens in blocks of 32, greedy, one seed, fp16, on one H200 GPU per task. C3a is compared with confidence top-$k$, Fast-dLLM's threshold (Wu et al., 2025), margin (Kim et al., 2025a), entropy-bounded (Ben-Hamu et al., 2025), KLASS (Kim et al., 2025b) and PC-Sampler (Huang et al., 2025) (full variant, background frequencies from FineWeb-Edu). Each decoder has one speed knob ($k, \tau, \gamma, \alpha, \ldots$), calibrated per task on fit-half problems only to 2, 4 and 8 tokens per step (speeds 1 to 3). Every knob is chosen by speed alone, never by accuracy. Speed is aggregate tokens per step, $256N/\sum \mathrm{NFE}$ over a configuration's $N$ problems. Table 3 shows speed 2 (about 4 tokens per step); Appendix A.4 has the sanity check and the speed curves. Each decoder is taken at its setting nearest to C3a's tokens per step, within $\pm 10\%$ (else —); the accuracy difference is paired on the shared problems, with a 95% percentile bootstrap interval from 2,000 resamples of problems. The comparison is exploratory and outside the C1 analysis; no multiple-testing correction is applied.

**Result.** At matched speed on GSM8K, C3a is indistinguishable in accuracy from the threshold and entropy-bounded decoders (C3a − threshold $-0.002$ $[-0.014, 0.012]$) and 0.053 to 0.082 more accurate than top-$k$, margin and PC-Sampler (Table 3): the gain is in how many tokens are committed per step, which a raw threshold already achieves. On MBPP (121 of 250 problems) it is again indistinguishable from both; the fixed-count decoders, over 10% faster, are not compared.

## 7 Related work

**Decoders.** LLaDA and Dream (Nie et al., 2025; Ye et al., 2025) scale masked diffusion (Sahoo et al., 2024) to 7–8B parameters and commit the most confident positions, as MaskGIT did (Chang et al., 2022); we audit that rule and propose no model. Other decoders replace top-1 probability (Kim et al., 2025a) with entropy, margin or stability (Ben-Hamu et al., 2025; Kim et al., 2025b), tune the cutoff (Wu et al., 2025; Shen and Ro, 2025) or learn the policy (Jazbec et al., 2026); they evaluate speed and accuracy, not the calibration of the tokens they commit.

**Calibration and the winner's curse.** Calibration is usually measured over all predictions (Guo et al., 2017). PC-Sampler (Huang et al., 2025) recalibrates the commit statistic for a positional bias; we add its weight as a covariate. Keeping the largest of noisy estimates biases it upward (Andrews et al., 2024), more so as the pool grows (Dalal et al., 2026), but leaves a correct posterior unbiased (Dawid, 1994; Efron, 2011). Diffusion LMs are overconfident against autoregressive models on most tasks (Yadav et al., 2026), an entropy-based commit order lowers their sequence-level calibration error (Jiang and Park, 2026), and commit confidence tracks correctness on math but not on factual recall (Asaria et al., 2026); none conditions on how many candidates a committed token beat.

**Dependency between tokens.** Dependency-aware decoders (Kim et al., 2026; Ringel et al., 2026; Azangulov et al., 2026) and ParallelBench (Kang et al., 2026) target term (A) of Eq. 1 and take the marginals as accurate. The split is not ours: EB-Sampler uses the same two terms (Ben-Hamu et al., 2025), and Tang et al. (2026) note that sampler and model error stay entangled; C2 computes both in nats where $p^*$ is known.

## Limitations

**Scope.** We audit two 8B-scale families and one 110M MDLM on FineWeb-Edu, GSM8K and MBPP, with model type and text type confounded; the 7–8B models' C1 traces are in fp16, as the GPUs we traced on lack bfloat16. C2's exact terms cover one synthetic distribution and one 110M fine-tune.

**Design and estimators.** C1 is teacher-forced: its states are those of an always-right decoder, not those after a free decoder's own mistakes, and a valid rewording counts as an error. The field term is only as context-aware as the state features of $\hat{\pi}(\ell, s)$: up to four, three in uniform states, where the step is constant.

**Contamination.** The screen flags only verbatim recall, and the time-split set mixes exposure with domain. Only the 110M model isolates the domain part. Our claims are about change with field size (E1, E2), not levels; the levels we report (E3, the native gap) are descriptive.

**Exploratory analyses and C3.** Analyses outside the E1/E2 Holm families (decoder states at $k \in \{2, 8\}$, matched uniform states, $\beta_z$, $\gamma_k$, E3, the native gap, the controls, and the robustness runs of §4.4) carry no claims. C3a is fitted on teacher-forced states but applied to free-running generation, and compared on one model, with greedy decoding, one seed and two tasks.

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# A Supplementary details

## A.1 C2: the main model and real-text details

**Main model.** One fine-tune masked the $\rho$ marker like any token (100M tokens); the main model never did, as at evaluation (300M). Neither moved the copied value's probability by our bar of 0.5 when the marker is switched from $\rho = 1$ to 0 with only the leader observed (main 0.251, other 0.014), so we took the lower validation NELBO (main 0.1751, other 0.1892 nats per token; the main model also trained on more tokens). The other crosses over at $k = 8$ for every $\rho \geq 0.5$ under both rules.

**Real text.** We skip documents under 256 tokens (fit-half ids up to 601; 600 for LLaDA) and keep at least 8 masked positions per state. Intervals resample documents (10,000 times).

## A.2 C1: E3, controls and committed tokens

**Weighting.** Averages are taken within a document, every state weighing the same, then across documents; positions are weighted by $1/(|M_s| \times$ states in their document) to match.

**E3, trajectory shift.** On the positions of decoder states at each $k$ and of matched uniform states (secondary: the uniform grid, reweighted to the decoder's rate mix), mask rate in $[0.05, 0.95)$, we fit $\Pr(a{=}1) = \sigma(\alpha_b + f(\ell) + \mathit{dec} \cdot (g_0 + h(\ell)))$, $\alpha_b$ per mask-rate bin (centred on $0.1, \ldots, 0.9$). The endpoint $\mathbb{E}_{\mathrm{dec}}[\pi_{\mathrm{ref}}(\ell, b) - \pi_{\mathrm{dec}}(\ell, b)]$ is positive if decoder-state tokens are right less often at equal confidence and mask rate; a sup-$t$ test checks $(g_0, h)$ jointly.

**Controls.** A change of $\Delta$ with $m$ could come from what winners are rather than from winning. *Position* (absolute position, distance to the nearest unmasked token, PC-Sampler's position weight; Huang et al., 2025) and *token type* (of the true token: punctuation, stop-word or content) are covariates $x$: $\mathbb{E}_m[\pi(\ell) - \pi(\ell, x)]$ is removed from $\Delta(m)$ and $x$ enters both field-term calibrators. *Temperature* fits $c = \sigma(\ell/T)$ per mask-rate bin on the fit half (a softmax temperature is a sensitivity row). Table 4 gives each control. For LLaDA-8B-Base in uniform states the change in E1 is below 0.0001 for position, 0.0029 [0.0027, 0.0032] for token type and 0.0005 [0.0004, 0.0005] for a temperature per mask-rate bin. In decoder states E1 keeps its sign under every control in every arm and moves by about 0.03 at most (temperature, LLaDA-8B-Instruct on MBPP: 0.156 to 0.125): the growth is not where winners sit, what kind of token they are, or an over- or underconfidence by mask rate. In uniform states the small base and 110M contrasts move by up to 0.004, so their sign is not robust. On the report half of the 198 documents of the 4-bit run (calibrators from their fit half), 4-bit NF4 in place of fp16 changes LLaDA-8B-Base's E1 by 0.0002 $[-0.0014, 0.0018]$ in uniform states and by 0.0055 $[-0.0004, 0.0115]$ in decoder states: both intervals hold 0, but in decoder states the difference is as large as that cell's E1 in Table 2, and fp16 E1 on these documents is itself 0.0003, so a precision effect there is not ruled out. Decoder-state E2 changes by $-0.0025$ $[-0.0030, -0.0021]$, of the size of the base models' E2.

**Committed tokens.** For tokens the $k{=}4$ decoder commits with 32–63 positions masked, $c - a$ is 0.079 [0.071, 0.086] for LLaDA-8B-Instruct on GSM8K, against an ECE of 0.033 over random masks; it is 0.013 to 0.018 for the base and 110M models and 0.079 to 0.176 for the instruct models, which reach 0.15 to 0.27 with 128–255 positions masked. Commit-time miscalibration has been reported for whole answers (Jiang and Park, 2026); here it holds per committed token. The mask count mixes field size with lost context, so this describes and does not test E1.

## A.3 Proofs of Propositions 1 and 2

Notation is as in §2, with $\ell_{-i} = (\ell_j)_{j \in M \setminus \{i\}}$, tie-break randomness $U$, and $W_i$ the event that $i \in P$ wins; $\pi(\ell_i, s)$ may depend on $i$ (suppressed).

**Proposition 1.** Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$. Given $s$, $(P, U)$ is independent of $(\ell_j, a_j)_{j \in M}$ and $(\ell_i, a_i)$ of $\ell_{-i}$, so the law of $(\ell_i, a_i, \ell_{-i}, P, U)$ given $s$ factorises as $p(P, U \mid s)\, p(\ell_i, a_i \mid s)\, p(\ell_{-i} \mid s)$, i.e. $(\ell_i, a_i) \perp Y \mid s$. By weak union $a_i \perp Y \mid Z$, so $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$ a.s. As $W_i$ is a function of $(Y, Z)$, conditioning on $(Y, Z)$ gives, for every bounded measurable $h$,

$$\mathbb{E}[a_i\, h(Z)\, \mathbf{1}_{W_i}] = \mathbb{E}[\pi(\ell_i, s)\, h(Z)\, \mathbf{1}_{W_i}], \qquad (6)$$

which is the defining property of $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s. on $W_i$. □

**Proposition 2.** Let $\mathbb{E}_m$ weight states by any function of $s$, let $(\ell, a)$ be the winner's pair, and take $\pi(\ell)$ as the calibration curve of the $m = 1$

| Arm | Masks | E1 | +pos. | +type | +temp. | E2 | +pos. | +type | +temp. |
|---|---|---|---|---|---|---|---|---|---|
| LLaDA Base | uniform | −0.003 | −0.003‡ | 0.000‡ | −0.002‡ | 0.001 | 0.001‡ | 0.003‡ | — |
| | dec. $k$=4 | 0.006 | 0.005‡ | 0.008‡ | 0.006‡ | 0.003 | −0.003‡ | 0.008‡ | — |
| Dream Base | uniform | −0.002 | −0.002 | −0.001‡ | −0.002‡ | 0.001 | 0.001‡ | 0.002‡ | — |
| | dec. $k$=4 | 0.015 | 0.015‡ | 0.021‡ | 0.016‡ | 0.002 | −0.001‡ | 0.008‡ | — |
| LLaDA-I GSM8K | uniform | −0.022 | −0.019‡ | −0.022‡ | −0.014‡ | −0.003 | −0.001‡ | −0.004‡ | — |
| | dec. $k$=4 | 0.024 | 0.022‡ | 0.023‡ | 0.028‡ | 0.021 | 0.015‡ | 0.022‡ | — |
| LLaDA-I MBPP | uniform | 0.025 | 0.022 | 0.024‡ | 0.014‡ | 0.013 | 0.037‡ | 0.012‡ | — |
| | dec. $k$=4 | 0.156 | 0.153 | 0.153‡ | 0.125‡ | 0.039 | 0.069‡ | 0.033‡ | — |
| Dream-I GSM8K | uniform | 0.061 | 0.061 | 0.059‡ | 0.075‡ | 0.014 | 0.035‡ | 0.012‡ | — |
| | dec. $k$=4 | 0.072 | 0.070‡ | 0.070‡ | 0.059‡ | 0.038 | 0.032‡ | 0.039‡ | — |
| Dream-I MBPP | uniform | 0.257 | 0.232‡ | 0.254 | 0.260 | 0.098 | 0.120‡ | 0.086‡ | — |
| | dec. $k$=4 | 0.186 | 0.168‡ | 0.183‡ | 0.163‡ | 0.077 | 0.078 | 0.075‡ | — |
| MDLM-110M | uniform | 0.002 | 0.002‡ | 0.006‡ | 0.000‡ | 0.002 | 0.002‡ | 0.005‡ | — |
| | dec. $k$=4 | 0.009 | 0.010‡ | 0.011‡ | 0.010‡ | −0.003 | −0.006‡ | 0.003‡ | — |

Table 4: Controls: E1 and E2 without a control and with position, token type, or a temperature per mask-rate bin (log-odds; it rescales confidence only, so has no E2). E2 here uses logistic calibrators for both $\hat{\pi}(\ell)$ and $\hat{\pi}(\ell, s)$, so with and without differ only by the control; its "without" value differs from the isotonic E2 of Table 2. ‡the paired with-minus-without difference has a 95% interval excluding 0. Exploratory.

population under the same weights. Exactly one $W_i$ holds per pool, so $a = \sum_{i \in M} a_i \mathbf{1}_{W_i}$ and $\pi(\ell, s) = \sum_{i \in M} \pi(\ell_i, s) \mathbf{1}_{W_i}$. Summing (6) with $h \equiv 1$ over $i$ under the law of $\mathbb{E}_m$ (it holds given each $s$, so for any state weights) gives $\mathbb{E}_m[a] = \mathbb{E}_m[\pi(\ell, s)]$. Hence $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$, and adding and subtracting $\pi(\ell)$ gives (3). At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property, $\pi$ being the calibration curve of this very population, and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ as just shown, so the field term is zero. □

The estimator inherits this last step only if $\hat{\pi}$ is fitted on the population the averages use. Ours are fitted on all states of an arm with pool weights (§3.3), while the averages keep states with at least 32 masked positions, so E2 is reported as a contrast.

### A.4 C3: sanity check, KLASS and speed curves (exploratory)

Before the comparison, the harness decodes 200 fit-half GSM8K problems at Fast-dLLM's settings (5-shot, 256 tokens, blocks of 32; fp16 on an RTX 2080 Ti). Top-$k$ at $k = 1$ scores 0.835 [0.777, 0.880] against a published 0.793, and the threshold rule at $\tau = 0.9$ 0.845 [0.788, 0.889] against 0.792 (Wu et al., 2025) (95% Wilson intervals): both published values lie inside, so the harness reproduces the two decoders it can be checked on, though our points are higher. Figure 2 plots accuracy against tokens per step at all three speeds; the cost per decoder and the speed-1 and speed-3 rows are in the code archive.

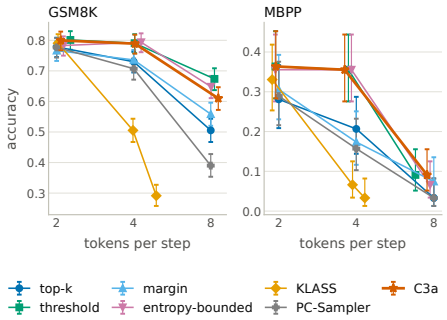



Figure 2: Accuracy against aggregate tokens per step, one panel per task (left GSM8K, right MBPP), one curve per decoder at its three speeds, with 95% Wilson intervals. Up and to the right is better.

**KLASS.** Ours differs from the reference (Kim et al., 2025b) in three ways, found after scoring: a KL history per 32-token block (two one-token steps at each block start), the reverse KL direction, and 32-token blocks, not 64 (as for all decoders). At its published thresholds it runs at about 2.6 tokens per step, as published; aiming at 4 and 8, the knob path, fixed in advance, lowers $\tau$ to 0.2–0.29 (reaching 3.9–4.9), far below the published range, partly to make up speed lost to the per-block history. Its speed-2 and speed-3 rows (†) describe that path, not KLASS as intended, and no comparison rests on them.